

🔥 XAI: Compiled Transformers as a Laboratory for Interpretability

FAMAF - UNC

First Semester 2026



eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

Compiled Transformers as a Laboratory for Interpretability ---

Compiled Transformers as a Laboratory for Interpretability

David Lindner, János Kramár, Matthew Rahtz, Tom McGrath, Vladimir Mikulik

DeepMind — April 2023

(Lindner et al. 2023)

🍷 The Core Problem: No Ground Truth ---



🍲 What If We Had Ground Truth? ---



🍷 Introducing Tracr: A Transformer Compiler for RASP ---



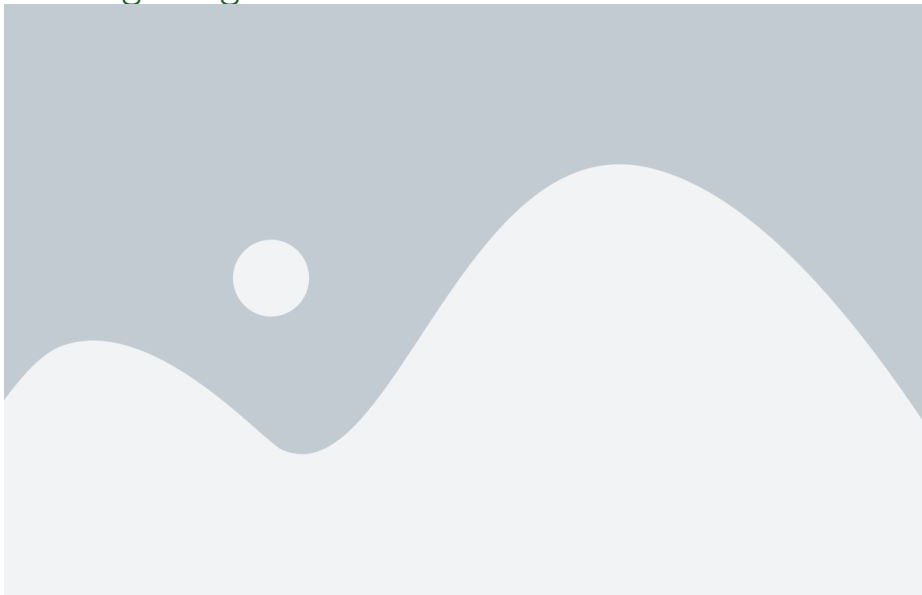
Known Mechanism (RASP program) $\longrightarrow$ Neural Network weights

🍷 Plan for Today ---



Part 1: Building the Compiler ---

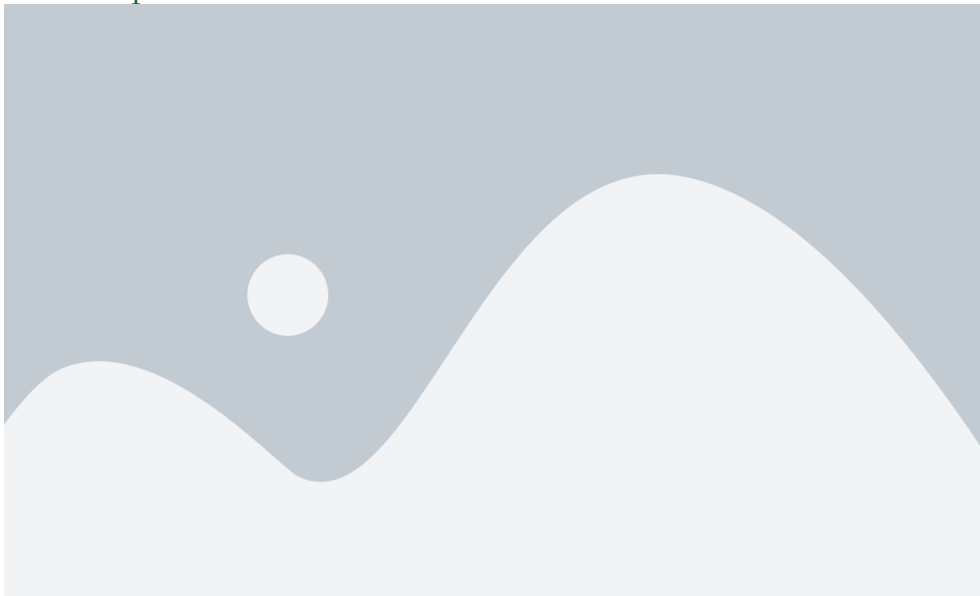
🍷 Hand-Coding Weights: Useful but Not Scalable ---



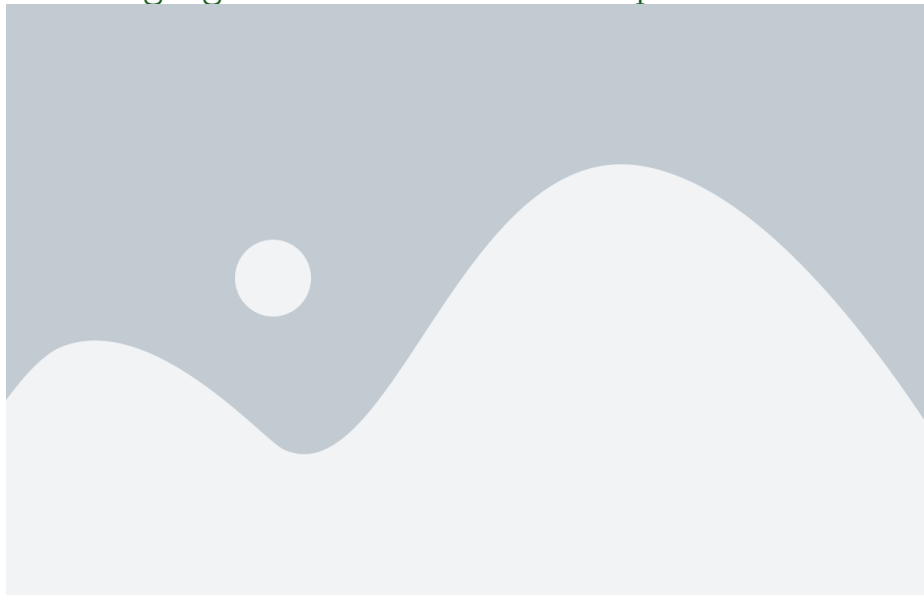
🍷 Tracr Analogous to a Compiler Pipeline ---



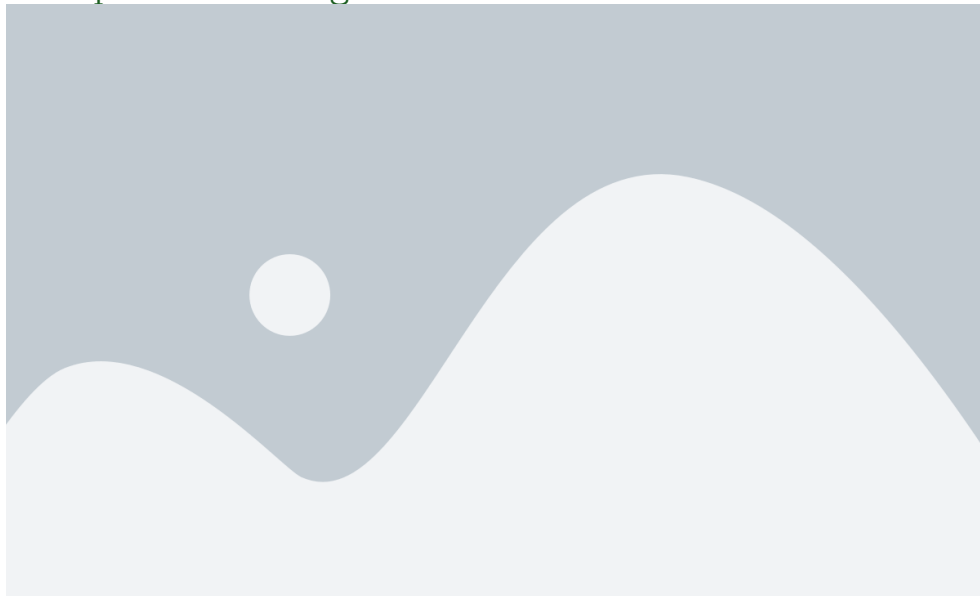
🍷 Three-Step Translation ---



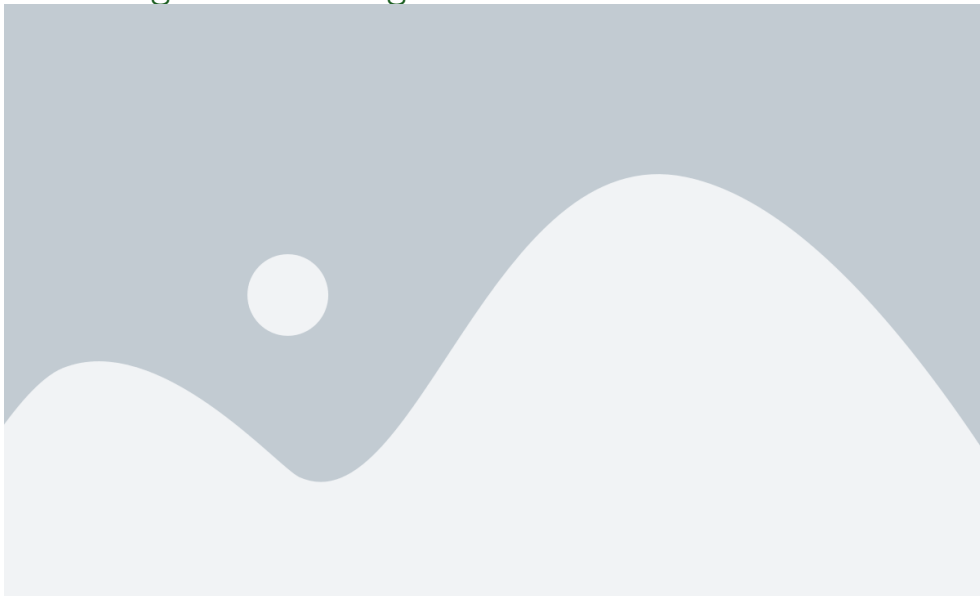
🍷 RASP: A Language for Transformer Computations ---



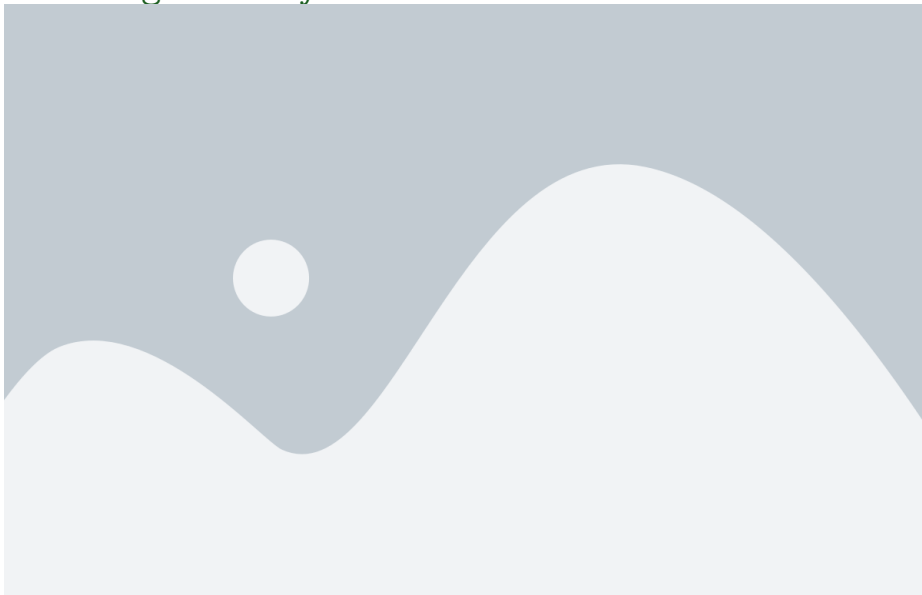
🍷 An Example RASP Program ---



🍷 Translating a RASP Program into a craft Transformer ---



🍷 Implementing MLP Layers ---



🍷 Implementing Attention Heads ---



🍷 craft Models Map to Any Transformer ---



🍷 Capabilities and Limitations of Tracr ---

Programs we can compile

- Count tokens and compute histograms
- Detect all occurrences of a pattern
- Sort the input sequence
- Check balanced parentheses (Dyck- n)

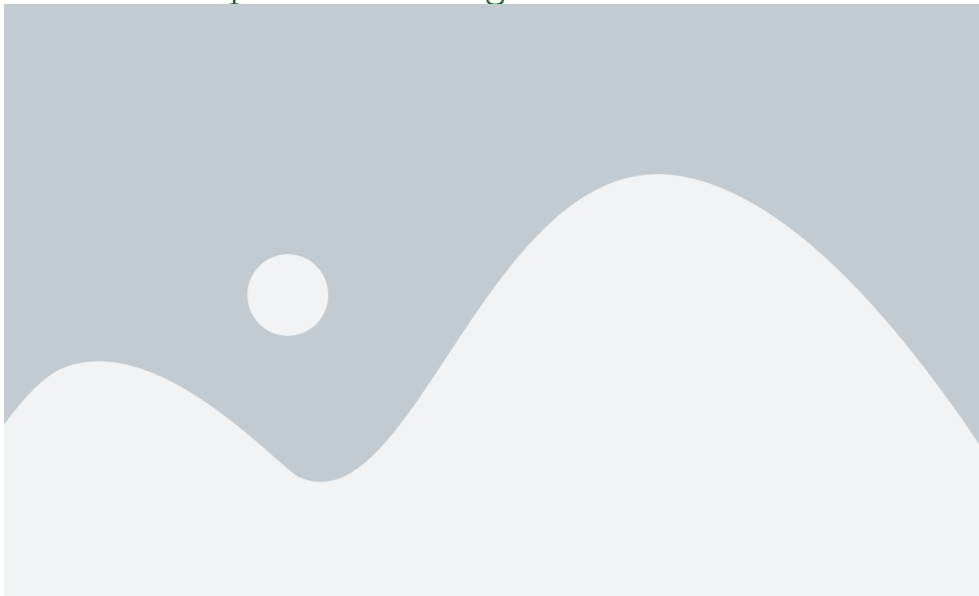
Limitations of RASP

- Binary attention patterns
- Designed for algorithmic, not probabilistic tasks
- Programs still close to transformer architecture

Limitations of Tracr

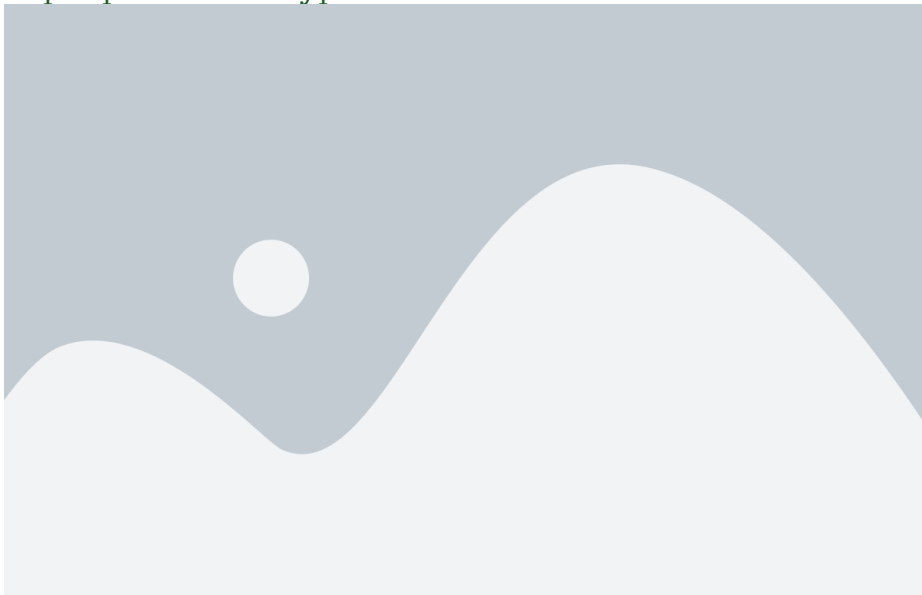
- Resulting models are large and inefficient
- Many possible optimizations missing
- Some advanced RASP features unsupported

🍷 We Can Now Compile RASP Programs! ---



🍲 Part 2: Studying Superposition in Compiled Models ---

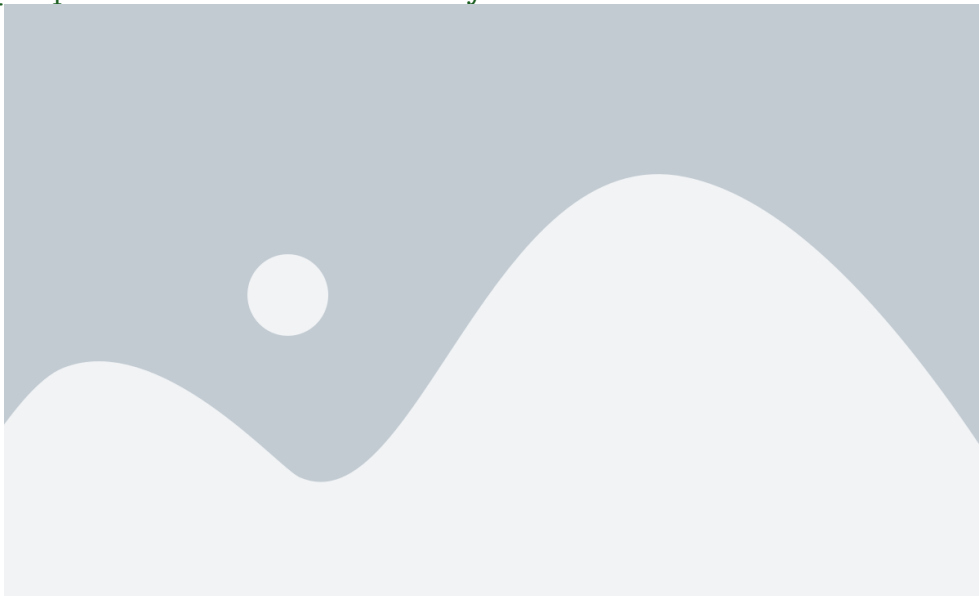
🍷 The Superposition Hypothesis ---



🍲 Superposition and Polysemanticity ---



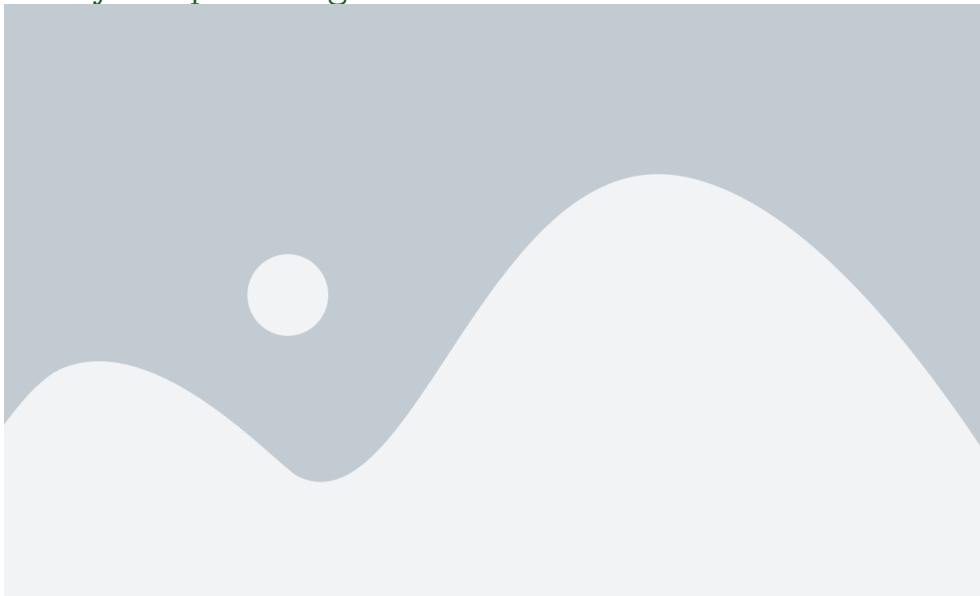
🍷 Superposition Occurs in Toy Models ---



🍲 Compressing Tracr Models to Induce Superposition ---



🍷 Linearly Compressing the Residual Stream ---



🍷 Embeddings Show Superposition ---



🍷 Which Features Are Stored in Superposition? ---



🍷 Future Directions for Tracr ---



Summary ---

- **Tracr**: RASP program → transformer weights with known mechanism
- Provides **ground truth** for evaluating interpretability methods
- Compiler pipeline: RASP → craft → neural network
- Compiled models **exhibit superposition** when residual stream is compressed
- Superposition follows toy-model predictions: importance, density, linear independence

Open source

<https://github.com/deepmind/tracr>

arXiv: 2301.05062

Thank You!

🍷 References --- |

- Elhage, Nelson, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, et al. 2022. “Toy Models of Superposition.” *Transformer Circuits Thread*.
- Lindner, David, János Kramár, Matthew Rahtz, Tom McGrath, and Vladimir Mikulik. 2023. “Compiled Transformers as a Laboratory for Interpretability.” *arXiv Preprint arXiv:2301.05062*.
- Weiss, Gail, Yoav Goldberg, and Eran Yahav. 2021. “Thinking Like Transformers.” In *International Conference on Machine Learning*, 11080–90. PMLR.